

A Meta Decision Tree Approach for B-cell Epitope Mining

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Abstract—The ability of antibodies to respond to an antigen depends on the antibodies’ specific recognition of epitopes, which are sites of the antigen to which antibodies bind. An increase in the availability of protein sequences and structures has enabled the identification of conformational epitopes, using various computational methods. The meta learner, among various approaches, has proved its feasibility and comparable accuracy in B-cell epitope prediction in previous studies. Nevertheless, its performance highly depends on the classification results of its multiple epitope base predictors within the meta learning architecture. We here propose bagging meta decision trees for epitope prediction to avoid the dependence on epitope prediction tools, and introduce 3D sphere-based attributes to improve prediction accuracy. Our experimental results demonstrate the superior performance of the bagging meta decision tree approach in comparison with single epitope predictors.

Keywords—B-cell; epitope; meta decision tree; ensemble learning

I. INTRODUCTION

The B-cell epitopes are specific regions on proteins that are recognized as the antigen binding sites of the antibodies of the B-cell receptors. A profound understanding of the interaction between antibodies and epitopes provides basis for diagnostics, therapeutics, and rational designs of preventive vaccines [1][2][3]. Therefore, generation of potent antibodies through reversed immunological approach requires precise knowledge of the epitope. Based on structure and interaction with the antibody, epitopes can be divided into two categories: conformational epitopes and linear epitopes. Roughly 10% of the B-cell epitopes are linear, and 90% are conformational [4][5]. A linear epitope is formed by a continuous sequence of amino acids, whereas a conformational epitope is composed of discontinuous sections of the antigen’s primary sequence. The discontinuous sections are close together in three-dimensional space, and interact with the antibody.

As the increase in the availability of protein structures, there have been more computational prediction tools developed to exploit various types of knowledge of protein structures. The knowledge of protein structures includes: (a) spacial neighborhood information, and a surface measure to make the prediction [6], (b) the ideas of unit patch of residue triangle and

the clustering coefficient to describe the local spacial context and compactness with two new parameters appended, ASA (Accessible Surface Area) propensity, and consolidated amino acid index [7], (c) the B-factor that reflects the fluctuation of atoms [8], etc. In addition, others adopted a hybrid approach that combine structural and physicochemical features [9], or proposed a meta learner that incorporates consensus results from multiple prediction servers by a voting mechanism [10], or through meta learning based on stacking and cascade generalization [11].

Though the previously proposed meta learning approach has successfully demonstrated its superior performance to that of other prediction tools, it suffers the dependence on the predictions of multiple epitope prediction tools. In this paper, we propose using a meta decision tree approach [12] for B-cell epitope prediction. Unlike the previous meta learners that rely on the classifications from some other B-cell epitope prediction tools that have been trained and may not be re-trained by the user from new data, the meta classifier we propose is independent of such kind of pre-trained B-cell epitope prediction servers. As a result, our approach is more applicable and flexible than the meta learning methods developed previously for B-cell epitope prediction. To evaluate the new B-cell epitope meta learner, we conducted the same cross-validation, and also did the comparison with several major epitope predictors based on the same independent test data set as used previously to keep the consistency. The results show that the proposed meta decision tree learner is not only comparable with the previous meta learners, but it also outperformed the other prediction servers markedly.

II. METHODS

A. Meta Decision Tree

Meta decision trees (MDTs) [12] belong to meta learning, which applies multiple base classifiers to a single data set by exploiting the classification results of the base classifiers as a sort of meta knowledge. In terms of the structure, a meta decision tree is identical to an ordinary decision tree in that both have internal nodes and leaves. What differentiate a meta decision tree from an ordinary decision tree are: the attributes

associated with the internal nodes, and the meaning indicated by the leaves.

In a decision tree, meta or ordinary, an internal node specifies a test on an attribute value. For an ordinary decision tree, the attribute selected for an internal node must be one of the base attributes used to describe the data instances, e.g. hydrophilic scale. By contrast, the attribute at an internal node of a meta decision tree is a meta attribute derived from the output of the base classifiers. It is worth noting that the base classifiers for a meta decision tree are standard inductive learners, e.g. an artificial neural network or a naïve Bayes classifier, not to confuse them with B-cell epitope prediction servers (e.g. BCPREDS [13]) used as the base predictors by other meta learning methods. Unlike those B-cell epitope prediction servers, e.g. BCPREDS, the base classifiers used in MDTs can be re-trained from new training data if necessary. As for the leaves, a leaf of an ordinary decision tree corresponds to a predicted class, but a leaf of a meta decision tree specifies a particular base classifier to use for class prediction. Fig. 1 illustrates an example of an ordinary decision tree and a meta decision tree, respectively. In Fig. 1(a), there are three internal nodes and four leaf nodes. Each internal node specifies a test on a particular base attribute values, e.g., $Feature_1 \leq 0.744$ (or > 0.744), and each leaf indicates the predicted class, e.g. C_1 . There are also three internal nodes, and four leaves in the meta decision tree, as shown in Fig. 1(b). Unlike the ordinary decision tree in Fig. 1(a), each internal node specifies a test on a particular meta attribute derived from the output of a base classifier, e.g., $entropy$ of SVM in Fig. 1(b), and instead of predicting the class, each leaf predicts which base classifier is most suitable for classification, e.g. C4.5. The advantage of an MDT is that it combines and exploits the classifications from multiple base classifiers to improve the accuracy of the final prediction.

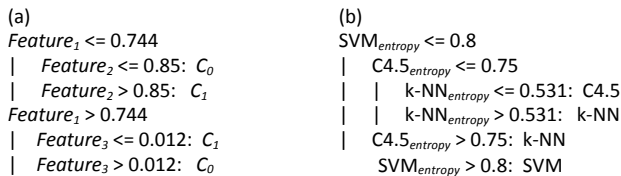


Fig. 1. Examples of ordinary decision trees and meta decision trees. (a) an ordinary decision tree, (b) a meta decision tree.

B. Base and Meta Attributes

Within the framework of machine learning, we translate the epitope prediction problem into an inductive learning problem. Given a training set of antigens with known epitope and non-epitope regions, the goal is to learn a classifier from a training set of antigens, and apply the learned classifier to novel antigens for epitope detection.

To describe each amino acid on a protein antigen, we first adopted 14 base attributes: epitope propensity, secondary structure, residue accessibility, B factor, solvent-excluded surfaces, solvent-accessible surfaces, protein chain flexibility, hydrophilicity, PSSM, atom volume, accessible surface area, side chain polarity, hydropathy index, and antigenic propensity. Descriptions and references of the base attributes

can be found in [11]. In addition, we also considered five other base attributes: surface probability [14], turns [15], accessible surface area defined by Lee and Richards [16], and two types of polarity defined by Ponnuswamy et al. [17] and Grantham [18], respectively. Furthermore, we extended the idea of Ansari and Raghava [19], and created the 3D sphere-based attributes from the base attributes. Unlike Ansari and Raghava, who only takes the amino acids in a linear window into account to generate 1D window-based attributes, we consider the amino acids in a constrained 3D spherical space, and analyze the values of different base attributes. To generate a 3D-based attribute value for an amino acid, we first create a surrounding sphere with its central carbon atom C_α as the center. We then compute the average value for a base attribute of all the amino acids within this 3D sphere, and use the average as the 3D sphere-based attribute value for the center amino acid. Because 90% of the B-cell epitopes are conformational (discontinuous), 3D sphere-based attributes are more informative than 1D window-based attributes. By varying the radius of the sphere, we can define different 3D sphere-based attributes. With the base attributes and the 3D sphere-based attributes, we represent the protein antigens in the training set, and train the base classifiers from the base training data set.

A meta attribute is defined over the output of the trained base classifiers. Currently, we use random forests (RF) [20], support vector machines (SVM) [21], C4.5 [22], k-NN [23], PART [24], BayesNet [25], JRip [26] and VotedPerceptron [27] as the base classifiers. Furthermore, the majority vote of the base classifiers is also included as a base classifier. Following Todorovski and Dzeroski [12], we calculate the properties of the class probability distributions predicted by the base classifiers that reflect the certainty and confidence of the predictions. In this study, we defined three meta attributes: $epi_prob(x, B)$, $entropy(x, B)$, and $vote_epi_prop(x)$, where x is a data instance, and B is a base classifier. The meta attribute $epi_prob(x, B)$ is the probability of epitope predicted by base classifier B for amino acid x . The meta attribute $entropy(x, B)$ is the entropy of the class probability distribution predicted by base classifier B for amino acid x . The meta attribute $vote_epi_prop(x)$ is the proportion of the epitope class predicted by all base classifiers for amino acid x . These meta attributes reflect the certainty of the base classifier in predicting the class, and they characterize the certainty differently.

We compute the meta attribute values for each data instance, i.e. amino acid, based on the output of the base classifiers, and combine them to form a meta training data set. We then train a meta decision tree from a training set of data that are described by the meta attributes.

C. Construction of Meta Decision Trees

The construction of a meta decision tree is identical to that of an ordinary decision tree. It is a greedy top-down, recursive search for the best decision tree from a training data set. The core algorithm is to select the best attribute for an internal node, and partition the data available at the node into subsets according to the attribute values to create the descending nodes. This process is repeated for the data associated with each descendant to select the next attribute to grow the tree until some stopping criterion is satisfied.

Instead of the commonly used measures of impurity reduction, such as information gain, gain ratio [22] or Gini [28], in ordinary decision trees, meta decision trees focus on the accuracy of each base classifier in predicting the data instances S available at an internal node. We define the new information measure as below:

$$info(S) = 1 - \max_{B \in Base} Accuracy(B, S) \quad (1)$$

where B is a base classifier, $Base$ is the set of all base classifiers, S is the data available at an internal node, and $Accuracy$ is the classification accuracy of B on S . We select the attribute that maximizes the decrease in $info$ of the subsets of S after the partition according to the values of the selected attribute as compared to the original $info$ of S . We repeat the same selection-partition process to grow the tree until the accuracy of some base classifier is 100% on the current subset, or the size of the current subset is below a pre-specified threshold. The classifier at the leaf with the maximum accuracy will be used to make the prediction for new instances after the tree has grown completely.

D. Bagging and Decision Threshold

Three factors affect the performance of a predictor: (a) training data size, (b) representation bias, and (c) search bias. Firstly, a learning algorithm for a predictor may find different hypotheses from a small data set, and each has the same prediction accuracy on the same training data. Nevertheless, not a single hypothesis is capable of covering the entire hypothesis space or a sufficiently large portion of the hypothesis space. Consequently, this can lead to the overfitting problem. Secondly, in most real-world applications, the true target concept may not be represented by any single hypothesis in the hypothesis space due to the limitations of representations. This constrains the applicability of the learned hypothesis. Thirdly, most learning algorithms adapt a local search strategy to avoid computational explosion during learning, e.g. the commonly used greedy partition rule for growing a decision tree; however, local search of any form can be stuck in local optima, and fail to identify the true target.

One approach for mitigating the aforementioned problems is ensemble learning. Various forms of ensemble learning have been developed. To further enhance the immunity of MDTs to the above problems, we apply the bagging strategy [29] to MDTs. By constructing an ensemble of MDTs from multiple bootstrap samples, we expect that the final prediction on the basis of the predictions of the ensemble of MDTs can further reduce the variance among different MDTs, and provide a better approximation to the true target.

Most standard learning algorithms assume balanced class distributions and equal misclassification cost, which limit their applicability to epitope prediction because the B-cell epitopes are severely underrepresented. Consequently, in addition to the ensemble learning (i.e. bagging and MDTs), we also consider imbalanced learning in this study, applying a validation-based cost-sensitive method to adjust the decision threshold. By considering the B-cell epitope prediction problem as a binary class prediction problem, i.e. epitope or non-epitope, we compute the class probability for an amino acid on the basis of

the classifications of the MDTs trained from the bootstrap samples. With m bagging MDTs, we define the class probability for an amino acid AA as follows:

$$P_c(AA) = \frac{1}{m} (\text{number of MDTs that predict Class } C) \quad (2)$$

where P_c is the probability of class C (i.e. C is epitope or non-epitope). To properly address the imbalanced class distribution in B-cell epitopes, we set a probability threshold for the final classification as follows:

$$C(AA) = \begin{cases} \text{non-epitope}, & P_{\text{epitope}}(AA) < T \\ \text{epitope}, & P_{\text{epitope}}(AA) \geq T \end{cases} \quad (3)$$

where T is a threshold. A carefully selected T based on cross-validation warrants a reasonable performance of the cost-sensitive bagging MDT approach. We show the entire system flow in Fig. 2.

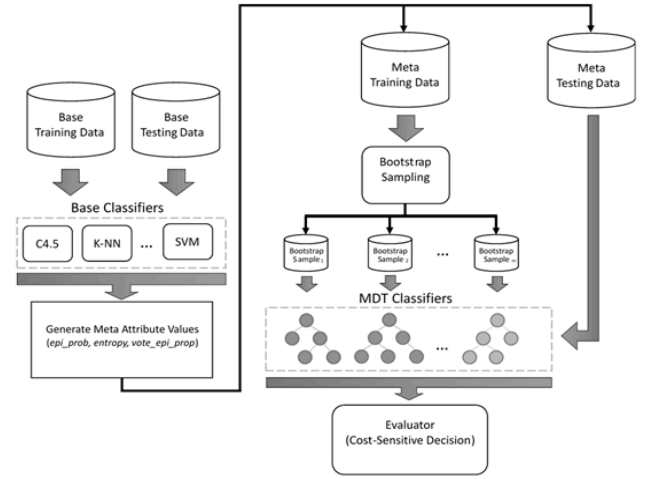


Fig. 2. The system flow of cost-sensitive bagging meta decision trees.

III. RESULTS

To evaluate the performance, we carried out two types of experiments: cross-validation and independent test. The performance measures we used include: TP rate (i.e., sensitivity), FP rate, precision (i.e., positive predictive value), percentage accuracy, F-score, and MCC (Matthews correlation coefficient). They are defined in Table 1.

TABLE I. DEFINITIONS OF PERFORMANCE MEASURES

Performance Measure	Definition
TPR ^a	$TP / (TP + FN)$
FPR	$FP / (FP + TN)$
Precision ^b	$TP / (TP + FP)$
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$
F-score	$2 \times TPR \times Precision / (TPR + Precision)$
MCC	$\frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$

^aTrue Positive Rate is also known as Sensitivity or Recall.

^bPrecision is also known as Positive Predictive Value.

From the data previously used to train the B-cell prediction servers, e.g. SEPPA [7], CBTOPE [19], and DiscoTope [30], etc., we collected 94 antigens for cross-validation. We first conducted a stratified 5-fold cross-validation (CV) to compare the bagging MDT approach with two other meta learners based on stacking and cascade, respectively [11]. Unlike the previous meta learners [10][11], the proposed bagging MDT approach does not rely on the predictions of other B-cell epitope prediction servers. Therefore, the unavailability of any prediction server does not affect the performance of bagging MDTs. This warrants the applicability and flexibility of our approach. It is also worth noting that to maintain the fairness and consistency in comparison, the 5-fold CV test did not include the B-cell prediction tools, e.g. SEPPA, because the data for the CV test were already used previously to train these tools during their development.

We randomly divided a data set of 94 antigens into five disjoint folds, each of approximately equal size. We stratified the folds to maintain the same distribution of epitopes and non-epitopes as in the original data set. We used one fold of data for testing prediction performance, and used the remaining four folds for training. We repeated the same training-testing process on each fold iteratively. Each run produced a result based on the fold selected for testing. The overall performance was used as the average of the results obtained from all iterations of the 5-fold CVs. We present the results in Table 2. The results show that bagging MDTs outperform the other meta learners markedly for MCC, F-score, and accuracy, and the paired t-tests also show a significant difference (p-value < 0.05).

TABLE II. RESULTS OF 5-FOLD CROSS-VALIDATION ON 94 PROTEIN ANTIGENS

Classifier	TPR	FPR	Precision	Accuracy	F-score	MCC	AUC
Meta Stacking	0.580	0.009	0.850	0.959	0.689	0.682	0.925
Meta Cascade	0.588	0.009	0.843	0.959	0.693	0.684	0.925
BaggingMDTs	0.751	0.018	0.784	0.964	0.767	0.748	0.955

In addition to the 5-fold cross-validation, we also evaluated the prediction performance based on an independent data set of 15 antigens after filtering out the antigens without epitope residues annotated in Epitope Information and B cell Assay Information in the IEDB, or previously used to train the B-cell epitope prediction servers. We compared bagging MDTs with twelve state-of-the-art B-cell epitope predictors: SEPPA [7], DiscoTope [30], CBTOPE [19], Bpredictor [31], ElliPro [32], AAP [33], ABCPred [34], BCPREDS [13], BepiPred [35], LBtope [36], Meta Stacking, and Meta Cascade [11]. We present the results in Table 3, and show a sample MDT in Fig. 3.

The results of the 5-fold CVs and the independent test indicate that the bagging MDT approach outperforms the current B-cell epitope prediction servers markedly for accuracy, F-score and MCC. In addition, the results also show that it is superior to the meta classifiers based on stacking or cascade, and its performance is independent of any B-cell epitope prediction server.

IV. CONCLUSION

Many different approaches exist for predicting linear and conformational B-cell epitopes. Among them one particular approach is based on meta learning that exploits the synergy among various prediction tools and demonstrates superior prediction performance in comparison with single epitope predictors. Nevertheless, the current meta learning approach to epitope prediction seriously depends on the predictive strength of other pre-trained conformational and linear epitope predictors that users cannot re-train directly. It limits the applicability and flexibility of meta classifiers. In this study, we propose a cost-sensitive bagging MDT approach. It combines two ensemble learning techniques with a cost-sensitive method. It does not employ the predictions of any pre-trained single epitope predictor to avoid the dependency. It applies the bagging mechanism to reduce the variance in the results of MDTs. It considers the misclassification cost to adjust the final prediction to address the imbalanced class distribution in B-cell epitopes. The experimental results of a 5-fold CV and an independent test demonstrate that the proposed meta-learning approach outperforms the single base tools and other recently developed meta-learning epitope predictors.

TABLE III. RESULTS OF INDEPENDENT TEST OF 15 PROTEIN ANTIGENS

Classifier	TPR	FPR	Precision	Accuracy	F-score	MCC	AUC
SEPPA 2.0	0.289	0.050	0.204	0.922	0.239	0.202	0.765
DiscoTope 2.0	0.930	0.763	0.051	0.266	0.097	0.080	0.699
Bpredictor	0.010	0.007	0.057	0.951	0.017	0.006	0.683
ElliPro	0.826	0.535	0.064	0.480	0.119	0.118	0.696
AAP	0.846	0.641	0.055	0.379	0.104	0.086	0.609
ABCPred	0.507	0.480	0.045	0.519	0.082	0.011	0.530
BCPREDS	0.990	0.874	0.048	0.163	0.091	0.072	0.570
BepiPred	0.761	0.499	0.063	0.512	0.117	0.106	0.656
LBtope	0.632	0.578	0.046	0.431	0.086	0.022	0.575
CBTOPE	0.159	0.003	0.681	0.961	0.258	0.317	0.681
Meta Stacking	0.194	0.008	0.520	0.958	0.283	0.300	0.793
Meta Cascade	0.199	0.008	0.519	0.958	0.288	0.304	0.789
BaggingMDTs	0.194	0.006	0.582	0.960	0.291	0.321	0.802

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vote_epi_prop(x) <= 0.375
|   entropy(x,SVM) <= 0.902552 : SVM
|   entropy(x,SVM) > 0.902552
|   |   epi_prob(x,PART) <= 0.083333 : VotedPerceptron
|   |   epi_prob(x,PART) > 0.083333
|   |   |   major_class_prop(x) <= 0.125
|   |   |   |   epi_prob(x,k-NN) <= 0.000023 : PART
|   |   |   |   epi_prob(x,k-NN) > 0.000023 : BayesNet
|   |   |   |   major_class_prop(x) > 0.125 : C4.5
|   |   |   major_class_prop(x) > 0.375 : Voting

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Fig. 3. A sample MDT.

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